

SUKHOBUZIMSKOYE, Russian Federation

WMO: 294770

Lat: 56.4971N

Lon: 93.2644E

Elev: 539

StdP: 14.41

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-48.2	-39.1	-51.7	0.3	-47.8	-44.3	0.4	-39.1	20.3	12.9	18.6	17.8	1.3	220	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	86.4	67.0	83.1	65.4	80.0	64.1	69.6	82.0	67.8	79.4	65.8	76.4	5.4	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.3	95.5	75.0	63.5	89.5	73.1	61.7	83.9	71.5	34.1	81.9	32.6	79.4	31.0	76.5	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.0	15.6	13.7	DB	-48.2	92.4	7.3	2.7	-53.5	94.4	-57.7	95.9	-61.8	97.5	-67.2	99.4	(4)
(5)			WB	-46.1	71.8	8.1	1.6	-51.9	73.0	-56.7	73.9	-61.2	74.8	-67.1	76.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.6	-5.4	2.8	18.8	37.5	48.8	62.5	66.0	61.7	47.0	32.7	11.5	4.8		
	DBStd	27.41	20.47	15.55	14.00	8.02	9.19	7.78	5.31	6.71	8.06	9.66	17.59	18.19		
	HDD50	7780	1716	1320	966	382	138	8	0	5	152	536	1156	1401		
	HDD65	12022	2181	1740	1431	825	508	132	52	142	539	1000	1606	1866		
	CDD50	1417	0	0	0	6	99	384	496	368	63	1	0	0		
	CDD65	186	0	0	0	0	4	59	83	40	0	0	0	0		
	CDH74	2391	0	0	0	5	103	898	857	498	30	0	0	0		
	CDH80	675	0	0	0	1	22	278	231	141	2	0	0	0		
	WSAvg	5.6	4.9	5.0	5.7	6.9	6.8	5.0	4.6	4.5	5.2	6.2	6.1	5.9		
	PrecAvg	15.2	0.5	0.5	0.5	0.9	1.5	2.0	1.9	2.7	1.7	1.2	1.0	0.9		
(15) Precipitation	PrecMax	18.7	1.0	0.9	1.2	1.6	2.9	3.2	3.9	4.1	2.9	2.1	1.8	1.9		
	PrecMin	11.5	0.1	0.2	0.1	0.2	0.3	0.7	0.1	1.6	0.4	0.3	0.5	0.4		
	PrecStd	2.0	0.3	0.2	0.3	0.4	0.8	0.8	0.9	0.7	0.7	0.5	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	36.3	48.3	70.6	84.6	91.4	90.0	89.6	78.4	61.2	49.2	37.3		
		MCWB	29.8	31.2	39.8	52.0	61.8	67.1	68.0	69.9	61.8	50.0	42.6	33.3		
	2%	DB	30.4	32.7	43.7	62.5	77.7	86.5	85.8	84.1	71.1	53.9	41.3	33.1		
		MCWB	27.0	28.7	36.8	48.1	58.3	66.8	67.3	67.9	59.1	45.7	36.5	30.0		
	5%	DB	26.6	29.8	40.4	57.4	71.8	82.8	82.4	79.7	66.1	49.5	37.9	30.1		
		MCWB	23.9	26.7	34.6	46.0	54.8	65.0	65.6	64.7	56.8	43.7	34.1	27.8		
	10%	DB	23.3	26.2	37.5	53.0	67.3	79.4	79.0	75.6	61.2	45.8	34.0	27.0		
		MCWB	21.1	23.6	32.4	42.8	53.6	63.5	64.7	62.7	53.4	41.3	31.3	24.8		
	0.4%	WB	30.6	31.7	40.6	53.3	63.5	71.1	71.1	71.2	62.5	50.5	42.9	33.8		
		MCDB	33.0	35.2	46.4	64.7	80.8	84.7	83.6	84.2	74.4	58.6	49.0	36.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	27.4	29.3	37.4	49.7	59.4	69.2	69.3	68.7	60.0	47.0	36.8	30.2		
		MCDB	30.3	32.4	42.7	61.7	73.1	82.8	80.5	80.9	70.0	52.3	40.6	32.7		
	5%	WB	24.1	26.6	35.0	46.5	56.7	66.9	67.8	66.3	57.2	44.1	34.2	27.9		
		MCDB	26.5	29.8	39.7	56.2	69.8	79.5	78.9	76.1	64.9	48.8	37.5	30.3		
	10%	WB	20.9	23.4	32.9	43.6	54.2	64.7	66.1	64.2	54.0	41.4	31.5	25.0		
		MCDB	23.2	26.3	37.1	52.2	65.6	76.6	76.2	73.5	60.5	45.5	34.0	26.9		
	5% DB	MDBR	16.8	19.8	20.7	18.9	21.1	22.8	19.9	20.6	18.4	14.4	14.6	15.9		
		MCDBR	16.5	18.2	21.1	29.1	30.5	31.3	26.8	28.5	29.1	22.0	12.0	16.8		
		MCWBR	14.6	15.3	15.4	16.8	15.3	13.6	11.2	13.2	17.3	14.6	8.9	15.0		
		MCWBR	15.6	17.4	18.6	26.5	26.6	26.6	22.8	24.5	27.8	20.4	11.9	16.2		
(35) Mean Daily Temperature Range	5% WB	MCWBR	13.9	15.0	13.9	16.1	14.9	12.2	11.2	12.6	17.1	14.0	9.4	14.7		
	taub		0.287	0.319	0.363	0.395	0.415	0.415	0.438	0.415	0.370	0.351	0.319	0.285		
		taud	2.011	1.991	1.954	2.131	2.188	2.280	2.225	2.256	2.336	2.231	2.056	1.960		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		177	222	243	258	261	263	253	250	246	213	161	139		
		Edh at Noon	21	32	43	42	42	39	41	37	30	25	20	16		
	RadAvg		237	548	981	1371	1559	1889	1700	1348	849	441	213	136		
(44) All-Sky Solar Radiation	RadStd		28	37	67	92	157	158	110	113	115	32	30	19		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page